

SPL-1 Project Proposal Form, 2026
Institute of Information Technology (IIT)
University of Dhaka

Student's Name:	Mst. Mehenaz Khatun		
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LogWarden: Log-Based Intrusion Detection System			
Project Description: The proposed project is an offline, command-line based Log-Based Intrusion Detection System designed to analyze server logs and detect suspicious activities. The system will process three types of logs: SSH authentication logs, web access logs, and outbound network/application logs. It will parse the different log formats and convert them into a common event representation for further analysis. The system will use rule-based detection techniques, counters, and sliding time windows to identify suspicious patterns such as repeated SSH login failures, successful logins following multiple failures, suspicious web resource probing, repeated web errors, and unusual outbound connections. It will also correlate related activities across different log types using common indicators such as source IP addresses, timestamps, usernames, and session information. Each detected incident will generate a severity-based alert along with the evidence that triggered it, while duplicate alerts will be suppressed within a configurable time period. The system will safely handle malformed, incomplete, duplicated, and out-of-order log records and will generate a final incident summary. Its detection performance will be evaluated using true positives, false positives, false negatives, precision, recall, processing time, and memory usage.			
Languages or Tools to be used: • C++ programming language • G++ compiler • Linux / WSL command line • Standard file handling			

- String and pattern matching / regular expressions
- Date and time handling
- Git and GitHub for version control

No SIEM product, IDS engine, log-analysis framework, or machine-learning library will be used.

Supervisor's Name: Prof. Dr. Md. Shariful Islam

Signature of the supervisor: _____

Date: _____

Before the Midterm Presentation:

I confirm that the progress is satisfactory and I am forwarding it for midterm presentation.

Signature of the supervisor: _____

Date: _____

Midterm Presentation Feedback:

Before the Final Presentation:

I confirm that the project is complete and I am forwarding it for final presentation.

Signature of the supervisor: _____

Date: _____